

ERAY KULKIZAGA

Backend Software Engineer | Python & Applied AI
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Python backend engineer focused on tested APIs, asynchronous processing, multi-tenant systems, and production-oriented AI workflows. Experienced with FastAPI, PostgreSQL, Redis, Docker, and LLM/RAG systems.

SKILLS

Languages: Python, SQL

Backend: FastAPI, Pydantic, SQLAlchemy, Alembic, REST APIs, WebSockets

Data & Infrastructure: PostgreSQL, Redis, Celery, pgvector

Cloud: AWS (ECS Fargate, RDS, ElastiCache, S3, ECR)

AI/ML: LLM APIs, RAG, TensorFlow/Keras, ONNX Runtime, OpenCV, MediaPipe

Engineering: Docker, GitHub Actions, pytest, Ruff, Pyright, Git

EXPERIENCE

FlyRank.ai | Backend AI Engineering Intern

Jul 2026 – Present

- Developed backend and AI engineering systems involving REST APIs, relational database integration, authentication, background processing, Docker, and LLM API integration.
- Built automation and data-processing workflows for responsible web scraping and automated PDF reporting with testing, validation, security controls, and technical documentation.
- Applied token- and cost-aware LLM integration, prompt evaluation, and workflow design while developing an end-to-end backend AI system.

Microsoft Türkiye | AI Innovators Summer Internship Program | [Live Demo](#)

Jun 2026 – Jul 2026

- Built QueryPilot Local, an offline-first PostgreSQL execution-plan assistant using FastAPI, Foundry Local, and RAG with read-only SQL validation and recursive EXPLAIN JSON analysis.
- Implemented deterministic diagnostics for missing indexes, expensive nested loops, disk-based sorts, cardinality misestimation, pg_stat_statements workload ranking, and plan-regression baselines.
- Achieved 12/12 rule-diagnosis and no-answer accuracy, 9/9 retrieval Hit@3, valid citations across evaluated AI outputs.

Iskur Holding | Enterprise Data Software Engineering Intern

Jun 2024 – Aug 2024

- Optimized SQL-based enterprise data retrieval using SAP Open SQL joins, reducing processing time by 41%.
- Developed backend validation and automation logic for contract, barcode, inventory, and operational-cost workflows.
- Implemented transactional logging and structured audit trails to improve monitoring, reporting, and traceability.

SELECTED PROJECTS

SignTurk | Real-Time Turkish Sign Language Recognition Platform | [GitHub](#) | [Demo Video](#)

Nov 2025 – Jun 2026

Certificate of Excellence — Outstanding Graduation Project

Tech: Python, FastAPI, WebSockets, TensorFlow/Keras, MediaPipe, ONNX Runtime, PostgreSQL, SQLAlchemy, React, OpenCV

- Co-developed an end-to-end real-time Turkish Sign Language platform; led model design, training, evaluation and FastAPI backend development.
- Trained a 226-class RTMW/RTMPose multi-stream ensemble on 28K+ AUTSL training videos, achieving 94.17% Top-1 and 99.49% Top-5 accuracy on the official signer-independent held-out test set.
- Engineered the live 179-class recognition pipeline using MediaPipe hand features, 16-frame motion sequences, BiLSTM inference, and real-time prediction streaming over WebSockets through FastAPI.

SupportFlow AI | Multi-Tenant AI Customer Support Platform | [Case Study](#)

Jul 2026 – Present

Tech: Python, FastAPI, PostgreSQL, Redis, Celery, Alembic, Docker, pytest, AWS ECS Fargate, LLM APIs, RAG, pgvector

- Co-developed a modular-monolith B2B platform with tenant-isolated authentication, RBAC, ticketing, messaging, membership management, and append-only audit workflows.
- Implemented transactional ticket, messaging, and document-ingestion pipelines with strict tenant isolation, atomic audit logging, status-transition rules, idempotent processing, and Redis-backed Celery worker recovery.
- Established Docker Compose and CI environments with 187 automated tests, Ruff, Pyright, Alembic migration checks, container-level smoke tests, and an AWS ECS Fargate deployment architecture using RDS, ElastiCache, S3, ECR, and GitHub OIDC.

EDUCATION

Eastern Mediterranean University | EMU — Famagusta, Northern Cyprus

Sep 2021 – Jun 2026

B.Sc. in Software Engineering

- ABET-accredited Software Engineering program
- English-medium instruction | Dean's High Honor List — 2 semesters

LANGUAGES

English | C2 (CEFR) | EF SET 4-Skill Certificate | [Verify Certificate](#)

Turkish | Native